

IPL CRUNCH'26

IPL DATA ANALYSIS **REPORT**

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PROJECT OVERVIEW

Ball by Ball match data to find important trends.
This project analysis IPL and performance patterns in T20 cricket. The analysis focus on:

- Toss impact on match outcome phase- wise scoring patterns
- Top performing batters and bowlers
- Match winning trends and strategic insights.

The dataset was cleaned and processed using python based data analysis techniques.

Visualisations were created different to understand better on how factors influence match results.

Tools used:

Python, Pandas, Matplotlib, Julius Ai

DATASET DESCRIPTION

The dataset used for this project contains detailed IPL Ball-by-Ball match transformation.

The data includes:

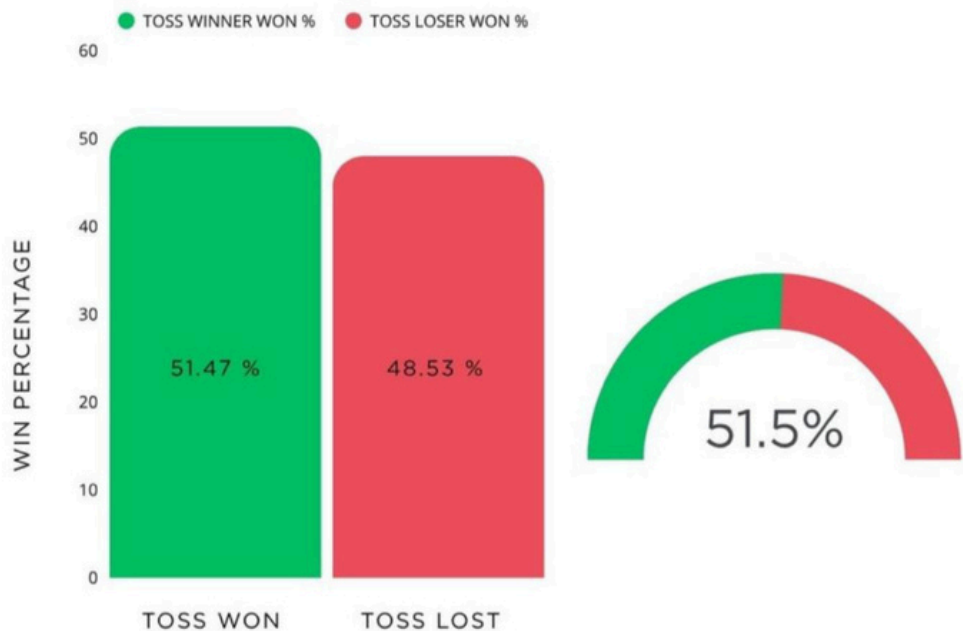
- Batting and Bowling teams
- Batter and Bowler statistics
- Runs scored on each delivery
- Wicket events
- Toss results
- Match winners
- Over-by-Over match progression.

The original dataset was provided in JSON format and was cleaned, transformed, and converted into a structured CSV Dataset using python and pandas for further analysis.

DATA PROCESSING PERFORMED

- Removed incomplete match records
- Standardised team and player names
- Handled missing values
- Converted overs into innings phases
- Structured nested JSON Data into
- tabular format.

TOSS IMPACT ON MATCH RESULTS

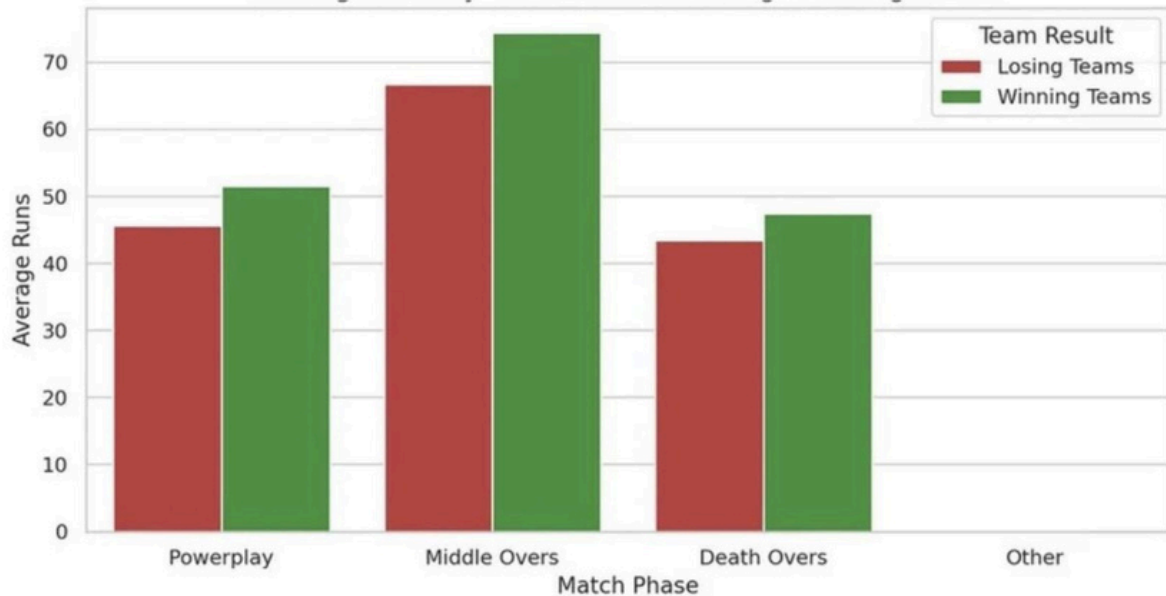


Out of completed matches:

- Toss winner won the match: 631 times that gives 51.47%
- Toss loser won the match: 595 times that gives 48.53%

Winning the toss does provide some advantage, but the edge is very small. A 51.47% win rate is only slightly above a pure 50-50 split, so the toss does not appear to be a strongly decisive factor across the dataset.

PHASE- WISE SCORING ANALYSIS



Phase	Winning teams	Losing teams	Gap
Powerplay overs	45.59	51.50	5.91
Middle overs	66.66	74.40	7.74
Death overs	43.43	47.40	3.96

The most important phase in this dataset is the **MIDDLE OVERTS**, that is the phase with the largest scoring gap between winning and losing teams. Winning teams score about 7.74 more runs than losing teams in overs 7-15. The power-play gap is also meaningful, but slightly smaller. The death overs matter too, though the difference there is lower than expected.

TOP PERFORMERS (2020- 2024)

Top 5 Batters by Total Runs (2020-2024)

Rank	Player Name	Runs	Seasons
1	Shubman Gill	2277	2020-2024
2	Faf du plessis	2269	2020-2024
3	Ruturaj Gaikwad	2176	2020-2024
4	Virat Kohli	2126	2020-2024
5	KL Rahul	2036	2020-2024

Top 5 Bowlers by Total Wickets (2020-2024)

Rank	Player Name	Wickets	Seasons
1	Harshal Patel	89	2020-2024
2	Yuzvendra Chahal	84	2020-2024
3	Rashid Khan	74	2020-2024
4	Avesh Khan	69	2020-2024
5	Mohammed shami	67	2020-2024

KEY INSIGHTS

When fans often credit power-play momentum or death- over fireworks for IPL wins, the data shows a more hidden truth: wicket preservation is a much stronger signal of victory. Teams posting 50k+ power-play runs, who won just 59.16%. This suggests that in the IPL, stability beats spectacle.

CONCLUSION

This project demonstrates how data analysis can uncover meaningful patterns and strategic insights in IPL cricket. The findings showed that toss advantage has a relatively small impact on match outcomes, while consistent scoring during the middle overs and wicket preservation are much more strongly associated with winning. Overall, the analysis highlights play a greater role in success than commonly assumed factors such as toss outcomes.